



JABchem



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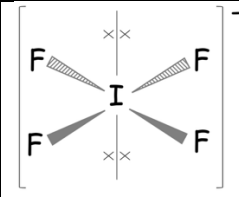
Past Papers Advanced Higher Chemistry

2026 Marking Scheme

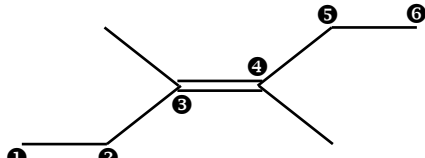
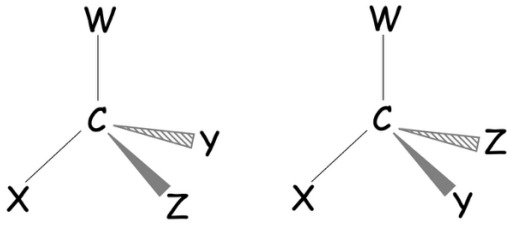
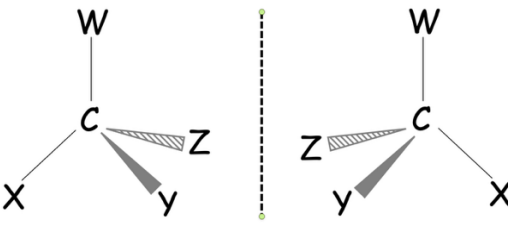
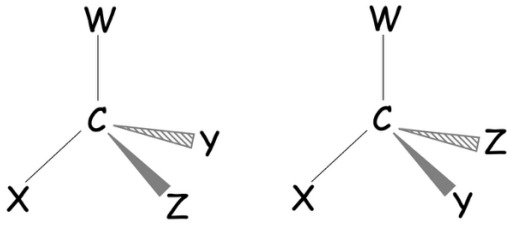
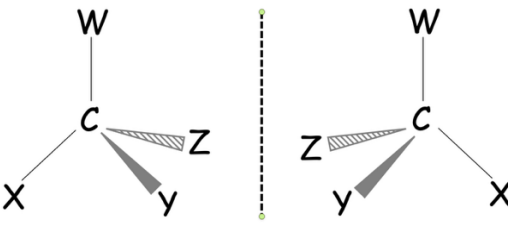
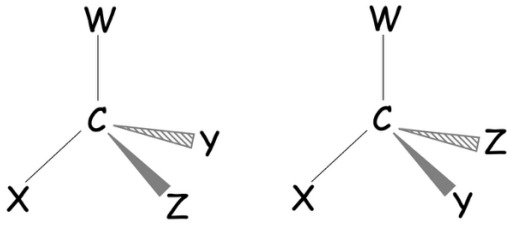
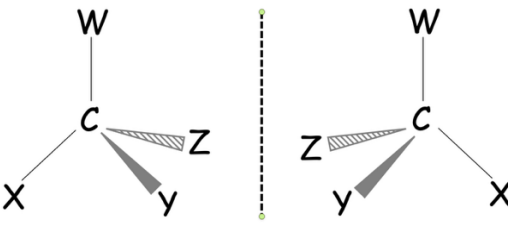
Grade Awarded	Mark Required		% candidates achieving grade
	(/160)	%	
A	108+	67.5%	%
B	90+	56.3%	%
C	73+	45.6%	%
D	55+	34.4%	%
No award	<55	<34.4%	%

Section:	Multiple Choice	Extended Answer	Project
Average Mark:	17.8 /27	46.4 /93	26.7 /40

2026 Adv Higher Chemistry Marking Scheme

MC Qu	Answer	% Correct	Reasoning
1	C	95	☒ A calcium is found in group 2 of the Periodic Table and is found in the s-block ☒ B carbon is found in group 6 of the Periodic Table and is found in the p-block ☒ C chromium is a transition metal in the Periodic Table and is found in the d-block ☒ D curium is found in the actinides row of the Periodic Table and is found in the f-block
2	D	69	Co atom is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$ and Co^{2+} ion is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$ ☒ A $3d^7$ has 2 pairs of electrons and 3 unpaired electrons (due to Hund's Rule) ☒ B as $4s^2$ has been removed in ionisation to Co^{2+} , 6 electrons in s orbitals ($1s^2$, $2s^2$ and $3s^2$) ☒ C Co^{2+} ion has 15 electrons in 3 rd main energy level ($3s^2 3p^6 3d^7$) ☒ D as 4 th shell electrons removed in Co^{2+} ion, electrons in 3 rd shell are highest energy
3	C	66	The Platinum in the complex is coordinated four ligands. Two NH_3 ligands and two OH_2 ligands. The two Cl^- ions are not in the complex because they are outside the square brackets.
4	D	61	Electron Pairs = $\frac{\text{No. of electrons in outer shell of central atom} + \text{No. of bonds} - \text{Charge}}{2}$ Electron Pairs = $\frac{7 + 4 - 1}{2}$ Electron Pairs = $\frac{12}{2} = 6$ pairs = octahedral electron pair arrangement  6 electron pairs form octahedral shape. - Lone pair to lone pair repulsion is greatest repulsion force so lone pairs take north and south pole position 180° apart - Four I-F bonds take up square planar arrangement in middle lone pair - lone pair repulsion > lone pair - bonding pair repulsion > bonding pair - bonding pair repulsion
5	A	29	$[Ca(OH)_2] = 0.25 \text{ mol l}^{-1} \therefore [OH^-] = 0.50 \text{ mol l}^{-1}$ $[H^+] = \frac{10^{-14}}{[OH^-]} = \frac{10^{-14}}{0.5} = 2 \times 10^{-14} \text{ mol l}^{-1}$ $pH = -\log_{10}[H^+] = -\log_{10}(2 \times 10^{-14}) = -(-13.7) = 13.7$
6	D	72	$NH_3 + NH_3 \rightleftharpoons NH_4^+ + NH_2^-$ acid base conjugate acid conjugate base
7	B	73	☒ A p-nitrophenol has range (5.4-6.0) $\therefore$ suitable for strong acid v weak alkali titrations ☒ B thymolphthalein has range (9.4-10.6) $\therefore$ suitable for weak acid v strong alkali titrations ☒ C bromophenol blue has range (3.0-4.6) $\therefore$ suitable for strong acid v weak alkali titrations ☒ D bromothymol blue has range (6.0-7.6) $\therefore$ suitable for strong acid v strong alkali titrations
8	B	63	☒ A If $\Delta G^\circ > 0$ then reaction is not feasible and no products would be formed ☒ B Reaction is feasible as $\Delta G^\circ < 0$ and $K > 1$ as more products than reactants at equilibrium ☒ C If $\Delta G^\circ > 0$ then reaction is not feasible and no products would be formed ☒ D If $K < 1$ then the equilibrium would lie to the left with more reactants than products
9	A	69 0	☒ A Increasing the temperature favours the reverse endothermic reaction and decreases K ☒ B Decreasing the temperature favours the forward exothermic reaction and increases K ☒ C Changes to pressure and concentration do not alter the value of the equilibrium constant ☒ D Changes to pressure and concentration do not alter the value of the equilibrium constant
10	C	77	The lower the value of K_a the weaker the acid (and stronger the base) ☒ A X is a weaker acid than Y as X has a lower K_a value ☒ B X is a weaker acid than Z as X has a lower K_a value $\therefore$ X is a stronger base than Z ☒ C Y is a weaker acid than Z as it has a lower K_a value ☒ D Y is a stronger acid than Z as it has a higher K_a value $\therefore$ Y is a weaker base than X

11	B	74	The lowest standard entropy would be the substance with the least disorder ($\therefore$ a solid not a liquid) ☒ A caesium has a mpt of 28°C and would be a liquid at 50° ☒ B calcium has a mpt of 842°C and would be a solid at 50° ☒ C mercury has a mpt of -39°C and would be a liquid at 50° ☒ D rubidium has a mpt of 39°C and would be a liquid at 50°																																																
12	A	71	☒ A First Order Kinetics as a doubling in [Q] gives doubling in reaction rate ☒ B This shape of graph does represent zero, first or second order kinetics ☒ C Zero Order Kinetics as a doubling in [Q] gives no change in reaction rate ☒ D Second Order Kinetics as a doubling in [Q] gives quadrupling in reaction rate																																																
13	C	69	☒ A Overall order of reaction = 3 (reactant A is 1st order and reactant B is 2nd order) ☒ B 1 molecule of A in the slow rate determining step $\therefore$ A is 1st order ☒ C 2 molecules of B in the slow rate determining step $\therefore$ B is 2nd order ☒ D X is not a reactant in the slow rate determining step so X is zero order																																																
14	C	74	<table><tr><th colspan="2">A (pentane)</th><th colspan="2">B (pent-1-ene)</th><th colspan="2">C (pent-1-yne)</th><th colspan="2">D (Cyclopentane)</th></tr><tr><th>Bonds</th><th>No of sigma bond</th><th>Bonds</th><th>No of sigma bond</th><th>Bonds</th><th>No of sigma bond</th><th>Bonds</th><th>No of sigma bond</th></tr><tr><td>5x C-C</td><td>5</td><td>4x C-C</td><td>4</td><td>4x C-C</td><td>4</td><td>5x C-C</td><td>5</td></tr><tr><td>12x C-H</td><td>12</td><td>10x C-H</td><td>10</td><td>8x C-H</td><td>8</td><td>10x C-H</td><td>10</td></tr><tr><td></td><td></td><td>1x C=C</td><td>1</td><td>1x C$\equiv$C</td><td>1</td><td></td><td></td></tr><tr><td></td><td>TOTAL = 17</td><td></td><td>TOTAL = 15</td><td></td><td>TOTAL = 13</td><td></td><td>TOTAL = 15</td></tr></table>	A (pentane)		B (pent-1-ene)		C (pent-1-yne)		D (Cyclopentane)		Bonds	No of sigma bond	Bonds	No of sigma bond	Bonds	No of sigma bond	Bonds	No of sigma bond	5x C-C	5	4x C-C	4	4x C-C	4	5x C-C	5	12x C-H	12	10x C-H	10	8x C-H	8	10x C-H	10			1x C=C	1	1x C $\equiv$ C	1				TOTAL = 17		TOTAL = 15		TOTAL = 13		TOTAL = 15
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15	B	66	<table><tr><th>Substance</th><th>Colour transmitted</th><th>Colour Absorbed</th><th>Wavelength Absorbed</th><th>Energy Absorbed</th><th>Size of Energy Gap between HOMO and LUMO</th></tr><tr><td>Lycopene</td><td>red</td><td>blue/green</td><td>490-500nm</td><td>lower energy</td><td>smaller gap</td></tr><tr><td>Vitamin A</td><td>yellow</td><td>blue</td><td>435-480nm</td><td>higher energy</td><td>larger gap</td></tr></table>	Substance	Colour transmitted	Colour Absorbed	Wavelength Absorbed	Energy Absorbed	Size of Energy Gap between HOMO and LUMO	Lycopene	red	blue/green	490-500nm	lower energy	smaller gap	Vitamin A	yellow	blue	435-480nm	higher energy	larger gap																														
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16	B	39	☒ A ethanol will react with carboxyl group in hydrocinnamic acid to form an ester ☒ B no C=C double bonds in hydrocinnamic acid to be decolourised by bromine solution ☒ C sodium hydroxide will react hydrocinnamic acid by neutralisation ☒ D concentrated sulfuric acid will react with benzene ring by electrophilic substitution																																																
17	A	36	Alcohol X must be a tertiary alcohol as it does not react with hot copper(II) oxide $\begin{array}{c} \text{H} \quad \text{CH}_3 \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{OH} \quad \text{H} \end{array}$ 2-methylpropan-2-ol $\xrightarrow[\text{conc H}_2\text{SO}_4]{\text{hot}}$ $\begin{array}{c} \text{H} \quad \text{CH}_3 \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}=\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \quad \quad \text{H} \end{array}$ methylpropene $\xrightarrow{\text{Br}_2}$ $\begin{array}{c} \text{H} \quad \text{CH}_3 \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{Br} \quad \text{Br} \quad \text{H} \end{array}$ 1,2-dibromo-2-methylpropane ☒ A $(\text{CH}_3)_3\text{COH}$ is 2-methylpropan-2-ol ☒ B $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$ is butan-2-ol ☒ C $\text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)_2\text{OH}$ is 2-methylbutan-2-ol ☒ D $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{OH}$ is 2-methylpropan-1-ol (primary alcohol so ruled out as the answer)																																																
18	C	71	☒ A as the chain length of the hydrocarbon group increases, the solubility decreases ☒ B as the chain length of the hydrocarbon group increases, the boiling point increases ☒ C the solubility decreases and boiling point increases as the chain length increases, ☒ D as the chain length of the hydrocarbon group increases, the solubility decreases																																																
19	A	72	$\begin{array}{c} \text{W} \\ \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{Cl} \quad \text{H} \end{array}$ 1 $\xrightarrow{\text{elimination}}$ $\begin{array}{c} \text{X} \\ \text{H} \quad \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}=\text{C} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$ 4 $\xrightarrow{\text{addition}}$ $\begin{array}{c} \text{Y} \\ \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{OH} \quad \text{H} \end{array}$ 2 $\xrightarrow{\text{oxidation}}$ $\begin{array}{c} \text{Z} \\ \text{H} \quad \text{O} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \quad \text{H} \end{array}$ 3																																																
20	B	73	$\begin{array}{c} \text{CH}_3\text{CH}_2\text{C}(\text{CH}_3)\text{CH}_2 \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{CH}_3 \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}=\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$ 2-methylbut-1-ene $\xrightarrow{\text{HCl}}$ $\begin{array}{c} \text{Major Product} \\ \text{H} \quad \text{H} \quad \text{CH}_3 \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{Cl} \quad \text{H} \end{array}$ 2-chloro-2-methylbutane or $\begin{array}{c} \text{Minor Product} \\ \text{H} \quad \text{H} \quad \text{CH}_3 \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \quad \text{Cl} \end{array}$ 1-chloro-2-methylbutane																																																

21	D	79	☒A the 2 carbon ethyl groups are on opposite sides of the C=C bond $\therefore$ <i>trans</i>- geometric isomer ☒B longest chain with C=C double bonds is 6 carbons so name ends in hex-3-ene ☒C the 2 carbon ethyl groups are on opposite sides of the C=C bond $\therefore$ <i>trans</i>- geometric isomer ☒D longest chain with C=C is hex-3-ene with two ethyl groups in <i>trans</i> position on carbons C₃ and C₄.  <i>trans</i>-3,4-dimethylhex-3-ene						
22	B	63	Methods to spot a enantiomer/optical isomer pair: <table><thead><tr><th>Method 1</th><th>Method 2</th></tr></thead><tbody><tr><td></td><td></td></tr><tr><td>Keep two corners the same and swap other two corners</td><td>Direct mirror image</td></tr></tbody></table>	Method 1	Method 2			Keep two corners the same and swap other two corners	Direct mirror image
Method 1	Method 2								
									
Keep two corners the same and swap other two corners	Direct mirror image								
23	A	59	There is only one hydrogen environment in cubane: the C-C single bonds with one hydrogen on each carbon						
24	C	56	no. of mol NaOH = volume $\times$ concentration = $0.05_{\text{litres}} \times 2_{\text{mol l}^{-1}} = 0.10_{\text{mol}}$ concentration = $\frac{\text{no. of mol}}{\text{volume}} = \frac{0.10_{\text{mol}}}{0.25_{\text{litres}}} = 0.4_{\text{mol l}^{-1}}$ NB Volume is 250cm^3 as 200cm^3 added to original 50cm^3						
25	B	68	Formula of paracetamol = $\text{C}_8\text{H}_9\text{O}_2\text{N}$ gfm $\text{C}_8\text{H}_9\text{O}_2\text{N} = (8 \times 12) + (9 \times 1) + (2 \times 16) + (1 \times 14) = 96 + 9 + 32 + 14 = 151$ $500\text{mg} = 0.5\text{g}$ paracetamol = 1 tablet 151g paracetamol = 1 tablet $\times \frac{151}{0.5} = 302$ tablets						

2026 Adv Higher Chemistry Marking Scheme

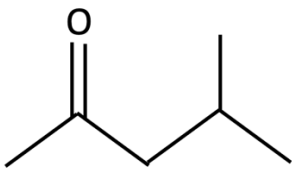
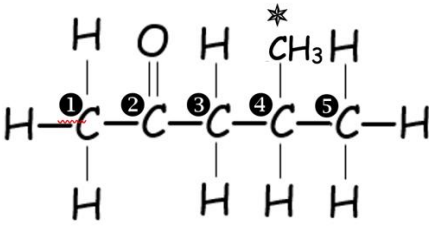
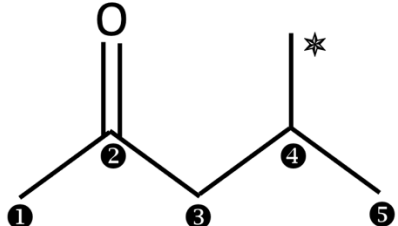
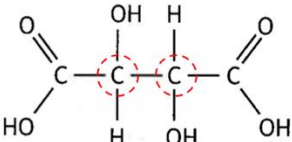
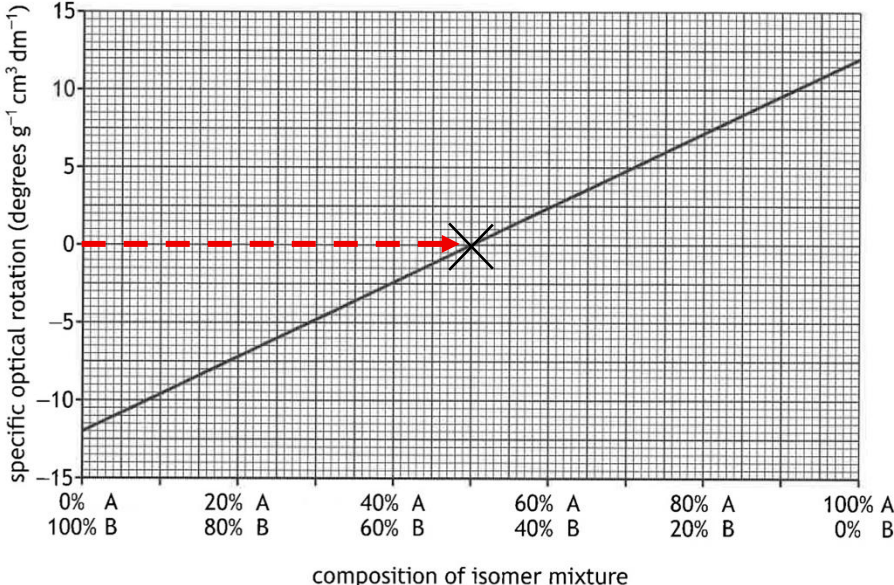
Long Qu	Answer	Reasoning																																																																																																																																									
1a	Orbital Box diagram showing:	↑↓ 1s ↑↓ 2s ↑↓ ↑↓ ↑↓ 2p ↑↓ 3s ↑↓ ↑↓ ↑↓ 3p ↑ 4s Potassium has 19 electrons and an electron arrangement of 2,8,8,1 The electrons fill up orbitals as per Aufbau principle, where 4s fills up before 3d.																																																																																																																																									
1b	0 m_l	Potassium has 19 electrons with an electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^4 4s^1$ For potassium, the outer electron is in the 4th shell <table><tr><th colspan="4">n = 4</th><th colspan="4">l = 0</th><th colspan="4">m_l = 0</th><th colspan="2">m_s = +$\frac{1}{2}$</th></tr><tr><th colspan="4">Principal Quantum Number is the shell number</th><th colspan="4">Angular Momentum Quantum Number is the shape of orbital</th><th colspan="4">Magnetic Quantum Number is the orientation of orbital</th><th colspan="2">Spin Quantum Number describes the spin direction of the electron</th></tr><tr><td>n=1</td><td>n=2</td><td>n=3</td><td>n=4</td><td>l=0</td><td>l=1</td><td>l=2</td><td>l=3</td><td>l</td><td>0</td><td>1</td><td>2</td><td>3</td><td colspan="2"></td></tr><tr><td>1st shell</td><td>2nd shell</td><td>3rd shell</td><td>4th shell</td><td>s</td><td>p</td><td>d</td><td>f</td><td></td><td></td><td></td><td></td><td></td><td colspan="2"></td></tr><tr><td colspan="4"></td><td colspan="4"></td><td>m_l</td><td>0</td><td>-1</td><td>-1</td><td>-1</td><td colspan="2"></td></tr><tr><td colspan="4"></td><td colspan="4"></td><td></td><td></td><td>0</td><td>0</td><td>0</td><td colspan="2"></td></tr><tr><td colspan="4"></td><td colspan="4"></td><td></td><td></td><td>+1</td><td>+1</td><td>+1</td><td colspan="2"></td></tr><tr><td colspan="4"></td><td colspan="4"></td><td></td><td></td><td></td><td>+2</td><td>+2</td><td colspan="2"></td></tr><tr><td colspan="4"></td><td colspan="4"></td><td></td><td></td><td></td><td></td><td>+3</td><td colspan="2"></td></tr></table> <table><tr><th colspan="2">m_s</th></tr><tr><td>+$\frac{1}{2}$</td><td>-$\frac{1}{2}$</td></tr></table>	n = 4				l = 0				m _l = 0				m _s = + $\frac{1}{2}$		Principal Quantum Number is the shell number				Angular Momentum Quantum Number is the shape of orbital				Magnetic Quantum Number is the orientation of orbital				Spin Quantum Number describes the spin direction of the electron		n=1	n=2	n=3	n=4	l=0	l=1	l=2	l=3	l	0	1	2	3			1 st shell	2 nd shell	3 rd shell	4 th shell	s	p	d	f																m _l	0	-1	-1	-1													0	0	0													+1	+1	+1														+2	+2															+3			m _s		+ $\frac{1}{2}$	- $\frac{1}{2}$
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1c(i)	Electrons are promoted to higher energy levels	Energy is absorbed which promotes electrons to higher energy levels in the atom. When these excited electrons fall back to a lower energy level then specific energy/wavelengths are releases and a line emission spectrum is created.																																																																																																																																									
1c(ii)	One answer from:	Different (electron) transitions/energy gaps. A suggestion that there are more than 2 energy levels/subshells. A suitable diagram with explanation.																																																																																																																																									
1c(iii)A	286	$E = \frac{L \times h \times c}{\lambda} \quad \therefore \lambda = \frac{L \times h \times c}{E}$ 1st Ionisation Energy Of K = 419 kJ mol⁻¹ $\lambda = \frac{L \times h \times c}{E} = \frac{6.02 \times 10^{23} \text{mol}^{-1} \times 6.63 \times 10^{-34} \text{ J s} \times 3 \times 10^8 \text{ m s}^{-1}}{419 \times 1000 \text{ J mol}^{-1}}$ $= 2.86 \times 10^{-7} \text{ m}$ $= 286 \times 10^{-9} \text{ m}$ $= 286 \text{ nm}$																																																																																																																																									
1c(iii)B	Shortest wavelength in Cs emission spectrum would have a longer wavelength	Caesium's outer shell is held with less force and has a lower 1 st ionisation energy value (376 kJ mol ⁻¹) as a result. As less energy is required to remove caesium's outer electron then the shortest wavelength in line emission spectrum would have a longer wavelength to reflect the lower energy required.																																																																																																																																									
2a(i)	107.8	$\Delta S^\circ = \Sigma S^\circ_{(\text{products})} - \Sigma S^\circ_{(\text{reactants})}$ $= (1 \times 76.0) + (\frac{1}{2} \times 205) - (1 \times 70.7)$ $= 76.0 + 102.5 - 70.7$ $= 107.8 \text{ J K}^{-1} \text{ mol}^{-1}$																																																																																																																																									
2a(ii)	841.4	$\Delta H^\circ = \Sigma \Delta H_f^\circ(\text{products}) - \Sigma \Delta H_f^\circ(\text{reactants})$ $= (1 \times 0) + (\frac{1}{2} \times 0) - (1 \times -90.7)$ $= 0 - -90.7$ $= 90.7 \text{ kJ mol}^{-1}$ $\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ = 0 \quad \therefore T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{90.7 \times 1000 \text{ J mol}^{-1}}{107.8 \text{ J K}^{-1} \text{ mol}^{-1}} = 841.4 \text{ K}$																																																																																																																																									
2b	0.072	1 minute 60 minutes = 1hour 8 hours 6 litres air 6x60 litres air 6x60x8 litres air = 2880 litres air 2.5x10⁻⁵ ppm = 2.5x10⁻⁵ mg Hg per 1 litre of air 1 litre air 2880 litres air 2.5x10⁻⁵ mg Hg vapour 2.5x10⁻⁵ mg Hg vapour x 2880 = 0.072 mq Hq vapour																																																																																																																																									

2c	Answer to include	1 st mark: Absorbance/intensity measured (at a specific wavelength) or The greater the concentration the greater the absorbance/intensity																											
		2 nd mark: Compare to standard/reference of known concentration or Using a calibration graph																											
3a(i)A	1 mark: Reaction 2 1 mark: is more likely as it has a more negative value of ΔG°	For Reaction 1: 2Ag + H ₂ S → Ag ₂ S + H ₂ ΔG _f ° = ΣΔG _f °(products) - ΣΔG _f °(reactants) = (1x-40.3) + (1x0) - (2x0) + (1x-33.0) = -40.3 - -33.0 = -7.3kJ mol ⁻¹ For Reaction 2: 2Ag + ½O ₂ + H ₂ S → Ag ₂ S + H ₂ O ΔG _f ° = ΣΔG _f °(products) - ΣΔG _f °(reactants) = (1x-40.3)+(1x-237) - (2x0)+(½x0)+(1x-33.0) = -40.3-237 - 0+0-33.0 = -277.3 - -33.0 = -244.3kJ mol ⁻¹																											
3a(i)B	Different temperatures will change the value of ΔG	ΔG° is carried out at standard conditions (298K and 1 atmosphere of pressure). When the temperature is changed then the different temperatures will give different values of ΔG due to the equation ΔG = ΔH-TΔS																											
3a(i)B	One answer from:	(ΔG is) not standard conditions ΔG° is determined under standard conditions A statement which recognises that ΔG is at different temperatures/pressures/concentrations to ΔG°.																											
3a(ii)A	Hexacyanidoferrate(II)	hexacyanidoferrate(II) = [Fe(CN)₆]⁴⁻ no. of ligands CN⁻ ligand metal ion Charge on complex charge on metal ion <table><tr><td colspan="2">Neutral ligands include:</td><td colspan="2">Negative Ligands include:</td><td>Central Ion:</td><td>Charge:</td></tr><tr><td>Ligand</td><td>Name</td><td>Ligand</td><td>Name</td><td>Positive Complex: metals keep their name</td><td rowspan="4">Charge of central ion is converted into roman numerals and put in brackets</td></tr><tr><td>H₂O</td><td>aqua</td><td>Chloride Cl⁻</td><td>chlorido</td><td>Negative Complex: Metals end in ATE</td></tr><tr><td>NH₃</td><td>ammine</td><td>Cyanide CN⁻</td><td>cyanido</td><td>e.g. Cuprate, Ferrate, Cobaltate</td></tr><tr><td>CO</td><td>carbonyl</td><td>Nitrite NO₂⁻</td><td>nitrito</td><td></td></tr></table> Iron ion has 2+ charge as Ion Complex has 4- charge overall as six cyanide CN⁻ ions contribute 6- charge.	Neutral ligands include:		Negative Ligands include:		Central Ion:	Charge:	Ligand	Name	Ligand	Name	Positive Complex: metals keep their name	Charge of central ion is converted into roman numerals and put in brackets	H ₂ O	aqua	Chloride Cl ⁻	chlorido	Negative Complex: Metals end in ATE	NH ₃	ammine	Cyanide CN ⁻	cyanido	e.g. Cuprate, Ferrate, Cobaltate	CO	carbonyl	Nitrite NO ₂ ⁻	nitrito	
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CO	carbonyl	Nitrite NO ₂ ⁻	nitrito																										
3a(ii)B	Answer to include:	1 st mark Electrons move to higher energy d orbitals 2 nd mark One answer from: <table><tr><td>blue absorbed</td><td>435 – 480 nm absorbed</td><td>complementary colour (to yellow) absorbed</td></tr></table>	blue absorbed	435 – 480 nm absorbed	complementary colour (to yellow) absorbed																								
blue absorbed	435 – 480 nm absorbed	complementary colour (to yellow) absorbed																											
3b(i)	0.00345 mol or 3.45x10 ⁻³ mol	Total no. of mol AgNO ₃ = volume x concentration = 0.05 x 0.100 = 0.005mol no. of moles SCN ⁻ ions = volume x concentration = 0.0137 x 0.0503 = 6.89x10 ⁻⁴ mol Ag⁺ + SCN⁻ → AgSCN 1mol 1mol 1mol 6.98x10⁻⁴mol 6.98x10⁻⁴mol 100cm ³ solution = 6.98x10 ⁻⁴ mol 500cm³ solution = 6.98x10 ⁻⁴ mol x 5 = 0.00345 mol ∴ no of mol Ag ⁺ excess ions = 0.00345 mol																											

3b(ii)	1.72%	$\therefore$ no of mol Ag^+ ions precipitated with Cl^- ions in cheese = $0.005\text{mol} - 0.00345\text{mol}$ = 0.00155mol $\begin{array}{ccccc} \text{Ag}^+ & + & \text{Cl}^- & \longrightarrow & \text{AgCl} \\ 1\text{mol} & & 1\text{mol} & & 1\text{mol} \\ 0.00155\text{mol} & & 0.00155\text{mol} & & \end{array}$ 0.00155mol Cl^- in cheese = 0.00155mol NaCl gfm $\text{NaCl} = (1 \times 23) + (1 \times 35.5) = 23 + 35.5 = 58.5\text{g}$ mass = no. of mol $\times$ gfm = $0.00155 \times 58.5 = 0.0907\text{g}$ $\% \text{ by mass} = \frac{\text{mass of NaCl}}{\text{mass of cheese}} = \frac{0.0907\text{g}}{5.27\text{g}} \times 100 = 1.72\%$											
4	Open Ended Question:	3 mark answer		2 mark answer	1 mark answer								
		Demonstrates a <u>good</u> understanding of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.		Demonstrates a <u>reasonable</u> understanding of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited</u> understanding of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.								
	<table><tr><td>0 marks</td><td>1 mark</td><td>2 marks</td><td>3 marks</td></tr><tr><td>48%</td><td>31%</td><td>16%</td><td>4%</td></tr></table>	0 marks	1 mark	2 marks	3 marks	48%	31%	16%	4%				
0 marks	1 mark	2 marks	3 marks										
48%	31%	16%	4%										
5a(i)A	compleximetric	EDTA is a compleximetric agent for volumetric titrations											
5a(i)B	hexadentate	EDTA has 6 positions on the molecule with non-bonding pairs of electrons which can be donated to the central metal ion.											
5a(i)C	Ligands donates non-bonding pair(s) of electrons to calcium ion	Ligands contain non-bonding (lone) pairs of electrons which provide both electrons to form a dative covalent bond with the central metal ion in a complex.											
5a(ii)A	one answer from:	available in high purity	be stable when solid and in solution	be soluble	reasonably high GFM								
5a(ii)B	To determine the accurate/exact concentration	EDTA (Ethylenediaminetetraacetic acid) will form a complex with some transition metal ions on a 1mol : 1mol basis. When the number of moles of EDTA is calculated the number of moles of ions can be worked out.											
5a(iii)	0.00269 or 2.69×10^{-3}	$\begin{array}{ccccc} \text{Ca}^{2+} & + & \text{EDTA}^{4-} & \longrightarrow & [\text{CaEDTA}]^{2-} \\ 1\text{mol} & & 1\text{mol} & & 1\text{mol} \\ 1.34 \times 10^{-4}\text{mol} & & 1.34 \times 10^{-4}\text{mol} & & \end{array}$ $\text{concentration} = \frac{\text{no. of mol}}{\text{volume}} = \frac{1.34 \times 10^{-4}\text{mol}}{0.05 \text{ litres}} = 0.00269 \text{ mol l}^{-1}$											
5b	247mg l^{-1} Hard	$1 \text{ mol Ca} = 40.1\text{g} \therefore 0.00269\text{mol Ca} \therefore 0.00269\text{mol CaCO}_3$ gfm $\text{CaCO}_3 = 100.1 \therefore \text{mass CaCO}_3 = 100.1 \times \frac{0.00269}{1} = 0.247\text{g} = 247\text{mg}$ $\therefore 0.00269\text{mol l}^{-1} = 247\text{mg l}^{-1}$ $\therefore$ Water hardness = hard (200 to 300 mg l^{-1})											
6a(i)	H_2	The first step in synthesis of an ether by nucleophilic substitution is to react a group 1 metal with an alcohol to make an alkoxide ion. Hydrogen is also produced. $\begin{array}{ccccccc} 2\text{CH}_3\text{CH}_2\text{OH} & + & 2\text{Na} & \longrightarrow & 2\text{CH}_3\text{CH}_2\text{ONa} & + & \text{H}_2 \\ \text{ethanol} & & \text{sodium} & & \text{sodium ethoxide} & & \text{hydrogen} \end{array}$											
6a(ii)	ethoxide	Alcohol	Methanol CH_3OH	Ethanol $\text{CH}_3\text{CH}_2\text{OH}$	Propan-1-ol $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$								
		Alkoxide Ion	Methoxide Ion CH_3O^-	Ethoxide Ion $\text{CH}_3\text{CH}_2\text{O}^-$	Propoxide Ion $\text{CH}_3\text{CH}_2\text{CH}_2\text{O}^-$								

6a(iii)	Diagram showing:	1 mark for curly arrows 1 mark for transition state									
6a(iv)	methoxyethane										
6b(i)	$1.05 \times 10^{-6} \text{ (mol l}^{-1} \text{ s}^{-1}\text{)}$	CH_3O^- is 1 st order due to CH_3O^- appearing to power of one in rate equation. Difference between Reaction 1 and Reaction 2 is concentration of $[\text{CH}_3\text{O}^-] \times 3$. The initial rate of reaction in Reaction 2 would be expected to be three times the initial rate of reaction for Reaction 1.									
6b(ii)	$0.000656 \text{ l mol}^{-1} \text{ s}^{-1}$ (1mark) (1mark)	$k = \frac{\text{rate}}{[\text{CH}_3\text{O}^-][\text{CH}_3\text{Cl}]} = \frac{3.15 \times 10^{-6} \text{ mol l}^{-1} \text{ s}^{-1}}{0.12 \text{ mol l}^{-1} \times 0.04 \text{ mol l}^{-1}} = 0.000656 \text{ l mol}^{-1} \text{ s}^{-1}$									
6b(iii)	58.2	$E_a = 2.303 \times R \times T \times \log_{10} \left(\frac{A}{k} \right)$ $E_a = 2.303 \times 8.314 \times 500 \times \log_{10} \left(\frac{718}{6.04 \times 10^{-4}} \right)$ $E_a = 2.303 \times 8.314 \times 500 \times \log_{10} (1.19 \times 10^6)$ $E_a = 2.303 \times 8.314 \times 500 \times 6.08$ $E_a = 58160 \text{ J mol}^{-1} = 58.2 \text{ kJ mol}^{-1}$									
7a(i)	reduction	Lithium Aluminium Hydride is a reducing agent. Alcohols oxidise into aldehydes & ketone so the reverse reaction is reduction.									
7a(ii)AI	One answer from:	(Carbon atom) has a partial positive (δ^+) charge (Carbon atom) is electron deficient (Carbon atom) is less electronegative than the O.									
7a(ii)AII	One answer from:	(Pi bond) requires less energy to break (than a sigma bond). (Pi bond) is higher in energy/less stable (than a sigma bond). (Pi bond) is weaker/easier to break (than a sigma bond)									
7a(ii)B	donates proton/ H^+	<table><tr><th>Term</th><th>Brønsted/Lowry Definition</th><th>Forms</th></tr><tr><td>Acid</td><td>Donates a proton/H^+ to the base</td><td>Conjugate base</td></tr><tr><td>Base</td><td>Accepts a proton/H^+ from the acid</td><td>Conjugate acid</td></tr></table>	Term	Brønsted/Lowry Definition	Forms	Acid	Donates a proton/ H^+ to the base	Conjugate base	Base	Accepts a proton/ H^+ from the acid	Conjugate acid
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7b		<table><tr><th>R₁ Component</th><th>R₂ Component</th></tr><tr><td>R₁ component is 3-methylbutan-1-ol as this alcohol must be a primary alcohol.</td><td>R₂ component is propan-2-ol this alcohol can be primary/secondary/tertiary.</td></tr></table>	R ₁ Component	R ₂ Component	R ₁ component is 3-methylbutan-1-ol as this alcohol must be a primary alcohol.	R ₂ component is propan-2-ol this alcohol can be primary/secondary/tertiary.					
R ₁ Component	R ₂ Component										
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7c(i)	2-methylpropan-1-ol	cyclic ether 2-methylpropan-2-ol 2-methylpropan-1-ol									
7c(ii)	One answer from:	<table><tr><td>Less steric hindrance.</td><td>Fewer/smaller atoms around the carbon.</td></tr><tr><td>More steric hindrance around other carbon atom.</td><td>Bulky groups around the other carbon atom.</td></tr><tr><td></td><td>More/larger atoms around the other carbon atom.</td></tr></table>	Less steric hindrance.	Fewer/smaller atoms around the carbon.	More steric hindrance around other carbon atom.	Bulky groups around the other carbon atom.		More/larger atoms around the other carbon atom.			
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8a(i)A	heating under reflux	A round-bottom flask with anti-bumping granules is heated and the condenser stops vapour escaping by condensing it back to a liquid which returns to the round-bottom flask. A heating mantle is used if flammable liquids are used.			
8a(i)B	anti-bumping granules	Anti-bumping granules (or glass beads) are added to give a surface for evaporation to take place on and prevent bumping (surge of liquids shooting up condenser)			
8a(ii)	Answer to include:	All 4 correct: 2 marks		2or3 correct: 1 mark	
		Minimum/small volume of solvent	Hot water/solvent	Cool	Filter
8a(iii)	0.47	gfm Eugenol = 164g no. of mol = $\frac{\text{mass}}{\text{gfm}} = \frac{3.16}{164} = 0.0193 \text{ mol}$ Eugenol $\longrightarrow$ Vanillin 1mol			

10b(i)		Full Structural Formula 	Skeletal Formula 
10b(ii)A	separating funnel	A separating funnel is a vessel to hold both solvents at same time with a tap at the bottom to allow accurate draining of the bottom layer and a stopper at the top to allow the separating funnel to be shaken thoroughly to allow transfer of solute from one solvent to another.	
10b(ii)BI	$K = \frac{[\text{tartaric acid}]_{\text{MIBK}}}{[\text{tartaric acid}]_{\text{aq}}}$	For reaction: $aA + bB \rightleftharpoons cC + dD$ $K = \frac{[C]^c [D]^d}{[A]^a [B]^b}$	
10b(ii)BI	0.00367 or 3.67×10^{-3}	$K = \frac{[\text{tartaric acid}]_{\text{MIBK}}}{[\text{tartaric acid}]_{\text{aq}}} \therefore [\text{tartaric acid}]_{\text{aq}} = \frac{[\text{tartaric acid}]_{\text{MIBK}}}{K} = \frac{0.0477}{0.649} = 0.0735 \text{ mol l}^{-1}$ no. of mol = volume $\times$ concentration = $0.05 \text{ litres} \times 0.0735 \text{ mol l}^{-1} = 0.00367 \text{ mol}$	
10b(ii)C	Multiple extracts using smaller volumes of solvent adding up to 50 cm^3	The quantity of solute extracted is greater if a number of extractions using smaller volumes of solvent are carried out rather than a single extraction using a large volume of solvent.	
10c(i)		A chiral centre is a carbon with four different groups attached to the carbon.	
10c(ii)A	X at zero rotation as shown:		

10c(ii)B	10.8°	specific optical rotation (degrees g⁻¹ cm³ dm⁻¹) composition of isomer mixture $[\alpha] = \frac{\alpha_{\text{observed}}}{c \times l} \therefore \alpha_{\text{observed}} = [\alpha] \times c \times l = 4 \times 1.35 \times 2 = 10.8^\circ$																									
11a(i)	active site	The active site of an enzyme is where the substrate fits perfectly into the shape of the active site allowing the catalysis of the chemical reaction.																									
11a(ii)	Covalent/sigma bond	For the bond to be so long lasting the weaker van der Waals forces as less likely and a stronger type of bonding must be responsible.																									
11b(i)	antagonist	<table><tr><th>Type</th><th>Definition</th></tr><tr><td>Agonist</td><td>An agonist binds to the receptor molecules to produce a response similar to the natural active compound.</td></tr><tr><td>Antagonist</td><td>An antagonist prevents the natural compound from binding to the receptor and so blocks the natural response from occurring.</td></tr></table>	Type	Definition	Agonist	An agonist binds to the receptor molecules to produce a response similar to the natural active compound.	Antagonist	An antagonist prevents the natural compound from binding to the receptor and so blocks the natural response from occurring.																			
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11b(ii)A	hydrolysis	Ester group Carboxyl group Hydroxyl group																									
11b(ii)B	C ₉ H ₁₀ O ₃	When converting skeletal formula to molecular formula, remember hydrogens are not drawn on benzene/phenyl groups or off the carbon chain. All carbons must be worked out to add to the formula.																									
12a	C ₄ N ₃ O ₂ H ₉	<table><tr><th>Element</th><th>C</th><th>N</th><th>O</th><th>H</th></tr><tr><td>%</td><td>36.6</td><td>32.1</td><td>24.4</td><td>6.9</td></tr><tr><td>No. of moles (divide % by gfm)</td><td>$\frac{36.6}{12} = 3.05$</td><td>$\frac{32.1}{14} = 2.29$</td><td>$\frac{24.4}{16} = 1.53$</td><td>$\frac{6.9}{1} = 6.90$</td></tr><tr><td>Mole ratio (divide through by smallest value)</td><td>$\frac{3.05}{1.53} = 1.99$</td><td>$\frac{2.29}{1.53} = 1.50$</td><td>$\frac{1.53}{1.53} = 1.00$</td><td>$\frac{6.90}{1.53} = 4.51$</td></tr><tr><td>Double to get rid of ½ and round to whole number</td><td>4</td><td>3</td><td>2</td><td>9</td></tr></table>	Element	C	N	O	H	%	36.6	32.1	24.4	6.9	No. of moles (divide % by gfm)	$\frac{36.6}{12} = 3.05$	$\frac{32.1}{14} = 2.29$	$\frac{24.4}{16} = 1.53$	$\frac{6.9}{1} = 6.90$	Mole ratio (divide through by smallest value)	$\frac{3.05}{1.53} = 1.99$	$\frac{2.29}{1.53} = 1.50$	$\frac{1.53}{1.53} = 1.00$	$\frac{6.90}{1.53} = 4.51$	Double to get rid of ½ and round to whole number	4	3	2	9
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12b(i)	131	gfm C ₄ H ₉ N ₃ O ₂ = (4×12)+(9×1)+(3×14)+(2×16) = 48+9+42+32 = 131. m=131 and z=1 ∴ m/z = 131/1 = 131																									
12b(ii)	[CH ₃] ⁺ methyl group	CH ₃ group has a mass of 15 and an m/z = 15																									
12c(i)	vibrate/stretch/bend	When infrared radiation is absorbed by organic compounds, bonds within the molecule vibrate/stretch/bend.																									

12c(ii)	<table><tr><td>C=O</td></tr><tr><td>C-H</td></tr><tr><td>C-N</td></tr></table>	C=O	C-H	C-N	<table><tr><th>Bond responsible for peak</th><th>Wavenumber (cm⁻¹)</th></tr><tr><td>O-H</td><td>2798</td></tr><tr><td>C=O</td><td>1700</td></tr><tr><td>C-H</td><td>1449</td></tr><tr><td>C-N</td><td>1242</td></tr></table>	Bond responsible for peak	Wavenumber (cm ⁻¹)	O-H	2798	C=O	1700	C-H	1449	C-N	1242
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12d	All singlets or No splitting or No muliplets	In high-resolution NMR, each hydrogen in a neighbouring environment adds a peak to the singlet generated. All peaks will be singlets if there. Are no neighbouring hydrogen environments.													
12e	Diagram Showing:	Remaining atoms to be allocated to creatine diagram: $C_4H_9N_3O_2 - CH_3N_3 = C_3H_6O_2$ 3 carbons make 12 bonds + 6 hydrogens make 6 bonds and 2 oxygens make 4 bonds makes a total of 22 bonds. - As there is O-H & C=O from IR spectrum suggests Carboxyl group - Carboxyl group -COOH would use up CHO_2 from remaining $C_3H_6O_2$ leaving C_2H_5 - Remaining C_2H_5 needs one hydrogen to attach to a nitrogen and C_2H_4 between nitrogen and carboxyl group 